

List of tools for static code analysis

This is a list of notable tools for static program analysis (program analysis is a synonym for code analysis).

Tool	Latest release	Free software	Languages supported						
			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>CodePeer</u>	2021-05-07 (21)	No; proprietary	Ada	—	—	—	—	—	—
<u>CodeScene</u>	2023-10-13 (6.3.5)	No; proprietary	—	C, C++, C#, Objective-C	Java, <u>Groovy</u> , <u>Scala</u>	JavaScript, TypeScript	VB.NET	Python	Swift, Go, PHP, Ruby
<u>CodeQL</u>	2026-02-05 (CLI: 2.24.1)	No; proprietary	—	C, C++, C#	Java, <u>Kotlin</u>	JavaScript, TypeScript	.NET	Python	Swift, Go, Ruby, Rust
<u>ConQAT (retired)</u>	2015-02-01	Yes; <u>ASL 2</u>	Ada	C#, C++	Java	JavaScript	—	—	ABAP
<u>Coverity</u>	2023-04-29 (2022.12) ^[5]	No; proprietary	—	C, C++, C#, Objective-C	Java	JavaScript, TypeScript	—	Python	Ruby, PHP
<u>ECLAIR</u>	2025-05-12 (3.14.0) ^[6]	No; proprietary	—	C, C++	—	—	—	—	—
Tool	Release	Free software	Languages supported						

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			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>CPAchecker</u>	2022-01-24 (2.1.1)	Yes; <u>ASL 2</u>	—	C	—	—	—	—	—
<u>Cppcheck</u>	2025-02-23 (2.17)	Partly; framework is GPL v3, with some proprietary features	—	C, C++	—	—	—	—	—
<u>Cppdepend</u>	2023-03-01 (2023.1) ^[7]	No; proprietary	—	C, C++	—	—	—	—	—
<u>Cpplint</u>	2020-07-29	Yes; <u>CC-BY-3.0</u> ^[8]	—	C++	—	—	—	—	—
<u>Fluctuat</u>	2001	No; proprietary	Ada95	C	—	—	—	—	—
<u>Fluid Attacks</u>	2026-01-07	No; proprietary	—	C#	Java, Kotlin, Scala	JavaScript, TypeScript	—	Python	PHP, Dart, Go, HCL, HTML, JSON, Ruby, Swift, XML, YAML
<u>Frama-C</u>	2022-06-21	Yes; <u>LGPL v2.1</u> , <u>BSD</u> , <u>QPL</u>	—	C	—	—	—	—	—
Tool	Release	Free software	Languages supported						

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			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>GrammaTech CodeSonar</u>	2020-06-01 (5.3)	No; proprietary	—	C, C++, Objective-C	Java	—	—	—	—
<u>GCC</u>	2023-4-26 (13.1)	Yes; GPLv3+ with GCC Runtime Library Exception	—	C	—	—	—	—	—
<u>HCL Security AppScan Source</u>	2020-12-01 (10.0.3)	No; proprietary	—	C, C++	Java, JSP	JavaScript	.NET	Python	ColdFusion, ASP, PHP, Perl, Visual Basic 6, PL/SQL, T-SQL, COBOL
<u>Helix QAC</u>	2023-04 (2023.1)	No; proprietary	—	C, C++	—	—	—	—	—
<u>Infer Static Analyzer</u>	2024-06-21 (1.2.0)	Yes; MIT	—	C, C++, Objective-C	Java	—	—	—	—
<u>Imagix 4D</u>	2020-10-01 (10.1.0)	No; proprietary	—	C, C++	Java	—	—	—	—
<u>Kiuwan</u>	2020-07-22	No; proprietary	—	C, C++, C#, Objective-C	Java, JSP	JavaScript	VB.NET	—	ABAP, COBOL, PHP, <u>PL/SQL</u> , <u>T-SQL</u> , SQL, Visual Basic, <u>Android</u>
<u>Klocwork</u>	2023-04-04 (2023.1)	No; proprietary	—	C, C++, C#	Java	JavaScript	—	Python	Kotlin
Tool	Release	Free software	Languages supported						

Tool	Latest release	Free software	Languages supported						
			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>Lint</u>	1978-07-26	Yes; permissive BSD-like ^[10]	—	C	—	—	—	—	—
<u>MALPAS</u>		No; proprietary	Ada	C	—	—	—	—	Pascal, Assembler (Intel, PowerPC and Motorola)
<u>Moose</u>	2021-01-21 (7.0.3)	Yes; MIT	—	C, C++	Java	—	.NET	—	Smalltalk
<u>NDepend</u>	2022-03-16 (2022.1) ^[11]	No; proprietary	—	C#	—	—	VB.NET .NET	—	—
<u>.NET Compiler Platform (Roslyn)</u>	2020-12-08 (3.8.0)	Yes; MIT	—	C#	—	—	VB.NET	—	—
<u>Parasoft C/C++test</u>	2024-12-03 (2024.2)	No; proprietary	—	C, C++	—	—	—	—	—
<u>Parasoft Jtest</u>	2024-12-03 (2024.2)	No; proprietary	—	—	Java	—	—	—	—
Tool	Release	Free software	Languages supported						

Tool	Latest release	Free software	Languages supported						
			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>PC-lint Plus</u>	2022-08-02 (2.0 Beta 2)	No; proprietary	—	C, C++	—	—	—	—	—
<u>PMD</u>	2025-11-28 (7.19.0)	Yes; BSD-like	—	C, C++	Java, Kotlin, Scala, JSP, <u>Apache Velocity</u> Template Language (VTL)	JavaScript	—	—	Apex, <u>ColdFusion</u> , HTML, PHP, PLSQL, Swift, Visualforce, XML, Maven POM, WSDL, XSL
<u>Polyspace</u>	2022-09-15	No; proprietary	Ada	C, C++	—	—	—	—	—
<u>Pretty Diff</u>	2019-04-21 (101.0.0)	Yes; <u>CC0</u>	—	—	—	JavaScript, TypeScript	—	—	Markup, script and style languages (like XML, CSS)
<u>PVS-Studio</u>	2024-08-16 (7.32)	No; proprietary	—	C, C++, <u>C++/CLI</u> , <u>C++/CX</u> , C#	Java	—	—	—	—
<u>Qodana</u>	2023-07-23 (2023.2)	No; proprietary	—	C#	Java, Kotlin	JavaScript, TypeScript	VB.NET	Python	Go, HTML, PHP, CSS, Android, Vue.js
<u>RIPS</u>	2020-02-17 (3.4)	No; proprietary	—	—	Java	—	—	—	PHP
<u>Semgrep</u>	2026-03-17 (1.156.0)	Yes; <u>LGPL v2.1</u>	—	—	Java	JavaScript, TypeScript	—	Python	Go, JSON, PHP, Ruby, language-
Tool	Release	Free software	Languages supported						

Tool	Latest release	Free software	Languages supported						
			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
									agnostic mode
<u>Sider</u>	2021-02-02	No; proprietary	—	—	—	JavaScript, CoffeeScript	—	Python	Ruby, PHP, Go
<u>SLAM project</u>	2010-07-14	No; proprietary	—	C	—	—	—	—	—
<u>SofCheck Inspector</u> , <u>Codepeer</u>	2020-08-24 (21.x)	No; proprietary	Ada	—	Java	—	—	—	—
<u>SonarQube</u>	2024-02-05 (10.4)	Partly; framework is <u>LGPL v3.0</u> , but some features can be proprietary	—	C, C#, C++, Objective-C	Java, <u>Kotlin</u> , Scala	JavaScript, TypeScript	VB.NET	Python	ABAP, Apex, CSS, COBOL, Flex, Go, HTML, PHP, PLI, PL/SQL, Ruby, Swift, TSQL, Visual Basic 6, XML
<u>SourceMeter</u>	2016-12-16 (8.2)	No; proprietary	—	C, C++	Java	—	—	Python	RPG IV (AS/400)
<u>Sourcetrail</u> (retired)	2021-04 (2021.4.19)	Yes; <u>GPL</u>	—	C, C++	Java	—	—	Python	Perl
<u>Sparse</u>	2021-09-06 (0.6.4)	Yes; <u>MIT</u>	—	C	—	—	—	—	GCC extensions
<u>Splint</u>	2007-07-12 (3.1.2)	Yes; <u>GPLv2</u>	—	C	—	—	—	—	—
Tool	Release	Free software	Languages supported						

Tool	Latest release	Free software	Languages supported						
			Ada	C, C++, C#, Objective-C	JVM	JavaScript, TypeScript	.NET, VB.NET	Python	Other languages
<u>StyleCop</u>	2016-05-02 (2016.1.0)	Yes; <u>Ms-PL</u>	—	C#	—	—	.NET	—	—
<u>Squore</u>	2020-11-27 (20.1)	No; proprietary	Ada	C, C++, C#, Objective-C	Java	JavaScript, TypeScript	VB.NET	Python	Fortran, PHP, PL/SQL, Swift, T-SQL, XAML
<u>Understand</u>	2023-01-19 (6.3)	No; proprietary	Ada	C, C++, C#, Objective-C	Java	JavaScript	—	Python	FORTRAN, Jovial, Pascal, VHDL, HTML, PHP, XML
<u>Visual Expert</u>	2021-09-10	No; proprietary	—	—	—	—	—	—	PowerBuilder, Oracle PL/SQL, SQL Server Transact-SQL (T-SQL)
<u>Visual Studio</u>	2021-10-12 (16.11)	No; proprietary	—	C, C++, C#	—	—	—	—	VB.NET
<u>Yasca</u> (retired)	2010-11-01 (2.21)	Yes; multiple licenses	—	C, C++	Java	JavaScript	—	—	ASP, PHP, HTML, CSS, ColdFusion, <u>COBOL</u>
Tool	Release	Free software	Languages supported						

Languages

Ada

- [CodePeer](#)
- [ConQAT](#)
- [Fluctuat](#)
- [LDRA Testbed](#)
- [MALPAS](#)
- [Polyspace](#)
- [SofCheck Inspector](#)
- [Squore](#)
- [Understand](#)

C, C++

- [Astree](#)
- [Axivion Suite \(Bauhaus\)](#)
- [BLAST](#)
- [Clang](#)
- [Coccinelle](#)
- [Coverity](#)
- [CPAchecker](#)
- [Cppcheck](#)
- [Cppdepend](#)
- [Cpplint](#)
- [ECLAIR](#)
- [Eclipse](#)
- [Fluctuat](#)
- [Frama-C](#)
- [GCC](#)
- [Helix QAC](#)
- [Facebook Infer](#)
- [Klocwork](#)
- [Lint](#)
- [LDRA Testbed](#)
- [Parasoft C/C++test](#)
- [PC-lint Plus](#)
- [Polyspace](#)
- [PVS-Studio](#)
- [SLAM project](#)
- [Sparse](#)
- [SonarQube](#)
- [Splint](#)
- [Understand](#)
- [Visual Studio](#)

C#

- [Axivion Suite \(Bauhaus\)](#)
- [Code Dx](#)
- [CodeScene](#)
- [CodeQL](#)
- [Coverity](#)
- [Fluid Attacks](#)
- [Kiuwan](#)

- [Klocwork](#)
- [.NET Compiler Platform](#)
- [PVS-Studio](#)
- [SonarQube](#)
- [Sotoarc](#)
- [StyleCop](#)
- [Squore](#)
- [Understand](#)
- [Visual Studio](#)

IEC 61131-3

- CODESYS Static Analysis – integrated add-on for [CODESYS](#) (application code realized e.g. in ST, FBD, LD)

Java

Tool	Latest release	Free software	Duplicate code	Notes
Checkstyle	2020-01-26	Yes; LGPL	No	Besides some static code analysis, it can be used to show violations of a configured coding standard. Duplicate code detection was removed ^[13] from Checkstyle.
Eclipse	2017-06-28	Yes; EPL	No	Cross-platform IDE with own set of several hundred code inspections available for analyzing code on-the-fly in the editor and bulk analysis of the whole project. Plugins for Checkstyle, FindBugs, and PMD.
FindBugs	2015-03-06	Yes; LGPL		Based on Jakarta BCEL from the University of Maryland. SpotBugs is the spiritual successor of FindBugs, carrying on from the point where it left off with support of its community.
IntelliJ IDEA	2021-04-06	Yes; ASL 2	Yes	A leading Java IDE with built-in code inspection and analysis. Plugins for Checkstyle, FindBugs, and PMD.
JArchitect	2017-06-11	No; proprietary		Simplifies managing a complex code base by analyzing and visualizing code dependencies, defining design rules, doing impact analysis, and by comparing different versions of the code.
Jtest	2024-11-01 (2024.2)	No; proprietary	Yes	Testing and static code analysis product by Parasoft .
Soot	2020-10-28	Yes; LGPL		A language manipulation and optimization framework consisting of intermediate languages.
PMD	2025-11-28	Yes; BSD License	Yes	Static code analyzer with support for plugins, including CPD . PMD supports checking of several languages.
Squale	2011-05-26	Yes; LGPL		A platform to manage software quality.
ThreadSafe	2014-03-28	No; proprietary		A static analysis tool focused on finding concurrency bugs.

- [Coverity](#)
- [Facebook Infer](#)
- [Fluid Attacks](#)
- [Klocwork](#)
- [LDRA Testbed](#)
- [PMD](#)
- [RIPS](#)
- [Semgrep](#)
- [SourceMeter](#)
- [Understand](#)

JavaScript

- [ESLint](#) – JavaScript syntax checker and formatter.
- [Google's Closure Compiler](#) – JavaScript optimizer that rewrites code to be faster and smaller, and checks use of native JavaScript functions.
- [CodeScene](#) – Behavioral analysis of code.
- [Fluid Attacks](#)
- [JSHint](#) – A community driven fork of JSLint.
- [JSLint](#) – JavaScript syntax checker and validator.
- [Klocwork](#)
- [Semgrep](#) – A static analysis tool that helps expressing code standards and surfacing bugs early. A CI service and a rule library is also available.
- [Understand](#)

Objective-C, Objective-C++

- [Clang](#) – The free Clang project includes a static analyzer. As of version 3.2, this analyzer is included in [Xcode](#).^[14]
- [Infer](#) – Developed by an engineering team at Facebook with open-source contributors. Targets null pointers, leaks, API usage and other lint checks. Available as open source on github.
- [Understand](#)

Opa

- [Opa](#) includes its own static analyzer. As the language is intended for web application development, the strongly statically typed compiler checks the validity of high-level types for web data, and prevents by default many vulnerabilities such as [XSS](#) attacks and database code injections.

Packaging

- [Lintian](#) – Checks Debian software packages for common inconsistencies and errors.
- [Rpmlint](#) – Checks for common problems in rpm packages.

Perl

- [Perl::Critic](#) – A tool to help enforce common Perl best practices. Most best practices are based on [Damian Conway's Perl Best Practices](#) book.
- [PerlTidy](#) – Program that acts as a syntax checker and tester/enforcer for coding practices in Perl.
- [Padre](#) – An IDE for Perl that also provides static code analysis to check for common beginner errors.

PL/SQL

- [TOAD](#) – A PL/SQL development environment with a Code xPert component that reports on general code efficiency as well as specific programming issues.
- [Visual Expert](#) – A PL/SQL code analysis tool^[15] that reports on programming issues and helps understand and maintain complex code ([Impact Analysis](#), [Source Code documentation](#), [Call trees](#), [CRUD matrix](#), etc.).

PowerBuilder, PowerScript

- [Visual Expert](#) – A tool scanning PowerBuilder libraries (PBLs) for [code inspection](#), [Impact Analysis](#), [Source Code documentation](#), [Call trees](#), [CRUD matrix](#).

Python

- [PyCharm](#) – Cross-platform Python IDE with code inspections available for analyzing code on-the-fly in the editor and bulk analysis of the whole project.
- [PyDev](#) – Eclipse-based Python IDE with code analysis available on-the-fly in the editor or at save time.
- [Pylint](#) – Static code analyzer. Quite stringent; includes many stylistic warnings as well.
- [Fluid Attacks](#)
- [Klocwork](#)
- [Semgrep](#) – Static code analyzer that helps expressing code standards and surfacing bugs early. A CI service and a rule library is also available.
- [Understand](#)

Transact-SQL

- [Visual Expert](#) – A SQLServer code analysis tool^[16] that reports on programming issues and helps understand and maintain complex code ([Impact Analysis](#), [source code documentation](#), [call trees](#), [CRUD matrix](#), etc.).

Tools with duplicate code detection

- [Axivion Suite](#) (Bauhaus)
- [Code Dx](#)
- [CodeScene](#)
- [PMD](#)
- [SofCheck Inspector](#)
- [SonarQube](#)

- [SourceMeter](#)
- [Understand](#)

Formal methods tools

Tools that use sound, i.e. over-approximating a rigorous model, formal methods approach to static analysis (e.g., using static program assertions). Sound methods contain no false negatives for bug-free programs, at least with regards to the idealized mathematical model they are based on (there is no "unconditional" soundness). Note that there is no guarantee they will report **all** bugs for buggy programs, they will report at least one.

- Astrée – finds all potential runtime errors by abstract interpretation, can prove the absence of runtime errors and can prove functional assertions; tailored towards safety-critical C code (e.g. avionics).
- CodePeer – Statically determines and documents pre- and post-conditions for Ada subprograms; statically checks preconditions at all call sites.
- ECLAIR – Uses formal methods-based static code analysis techniques such as abstract interpretation and model checking combined with constraint satisfaction techniques to detect or prove the absence of certain run time errors in source code.
- ESC/Java and ESC/Java2 – Based on Java Modeling Language, an enriched version of Java
- Frama-C – An open-source analysis framework for C, based on the ANSI/ISO C Specification Language (ACSL). Its main techniques include abstract interpretation, deductive verification and runtime monitoring.
- KeY – analysis platform for Java based on theorem proving with specifications in the Java Modeling Language; can generate test cases as counterexamples; stand-alone GUI or Eclipse integration
- MALPAS – A formal methods tool that uses directed graphs and regular algebra to prove that software under analysis correctly meets its mathematical specification.
- Polyspace – Uses abstract interpretation, a formal methods based technique,^[17] to detect and prove the absence of certain run time errors in source code for C/C++, and Ada
- SPARK Toolset including the SPARK Examiner – Based on the SPARK language, a subset of Ada.

See also

- [Automated code review](#)
- [Best Coding Practices](#)
- [List of software development philosophies](#)
- [Dynamic program analysis](#)
- [Software metrics](#)
- [Integrated development environment \(IDE\)](#) and [comparison of integrated development environments](#). IDEs will usually come with built-in support for static program analysis, or with an option to integrate such support. Eclipse offers such integration mechanism for most different types of extensions (plug-ins).

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External links

- The Web Application Security Consortium's Static Code Analysis Tool List (<http://projects.webappsec.org/w/page/61622133/StaticCodeAnalysisList>)
 - SAMATE-Source Code Security Analyzers (http://samate.nist.gov/index.php/Source_Code_Security_Analyzers.html)
 - SATE – Static Analysis Tool Exposition (<http://samate.nist.gov/SATE.html>)
 - "A Comparison of Bug Finding Tools for Java" (<http://www.cs.umd.edu/~jfooster/papers/issre04.pdf>), by Nick Rutar, Christian Almazan, and Jeff Foster, University of Maryland. Compares Bandera, ESC/Java 2, FindBugs, JLint, and PMD.
 - "Mini-review of Java Bug Finders" (http://www.oreillynet.com/digitalmedia/blog/2004/03/minireview_of_java_bug_finders.html), by Rick Jelliffe, O'Reilly Media.
 - "Static Code Analyzer" (<http://derscanner.com>)
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